

Chao Chang

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Education: PhD student

Major: Environmental sciences

Date of birth: 1995.02

Living: Shaanxi, China

Research interests: Aquatic ecology; Community ecology; eDNA; Biogeochemistry



Education

2022.09-present	Northwest A&F university	Environmental Sciences	PhD student
2019.09-2022.06	Northwest A&F university	Environmental Sciences	Master
2014.09-2018.06	North University of China	Environmental Engineering	Bachelor

Publication list

(a) First author

1. **Chang, C.**, Zhou, Y., Zhang, C., Zhang, Q., Ye, S., & Li, M.* (2026). Environmental Stress Drives Biotic Homogenization and Food-Web Destabilization Across River Basins with Contrasting Pollution. *Environmental Science & Technology*, 60(7), 5270-5284. (IF=11.3, JCR Q1)
2. **Chang, C.**, Wu, J., Wang, Y., Ye, S., Ren, M., Ma, Y., & Li, M.* (2026). Environmental Stress-Driven Trophic Associations Modulate Fish Biogeography in a Multi-Tributary River Basin. *Molecular Ecology*, 35(4), e70268. (IF=3.9, JCR Q1)
3. **Chang, C.**, Ren, M., Wang, H., Ye, S., Tang, X., He, D., Hu, E., & Li, M.* (2025). Riverine network size determined major driving factors of the composition and diversity of aquatic invertebrate communities in a multi-tributary mountain river basin. *Water Research*, 123257. (IF=12.4, JCR Q1)
4. **Chang, C.**, Ma, Y., Ye, S., Tang, X., Ren, M., Hu, E., & Li, M.* (2025). Network-scale spatiotemporal dynamics and drivers of nitrogen-cycling genes and their microbial hosts in a multi-tributary mountain river. *Water Research*, 125211. (IF=12.4, JCR Q1)
5. **Chang, C.**, Hu, E., Tang, X., Ye, S., Zhao, D., Qu, Z., & Li, M.* (2025). Assembly of soil multitrophic community regulates multifunctionality via multifaceted biotic factors in subtropical ecosystems. *Environment International*, 195, 109272. (IF=9.7, JCR Q1)
6. **Chang, C.**, Hu, E., Xue, X., Li, J., Du, D., Yang, F., & Li, M.* (2024). Hydro-morphology and water quality jointly shape the structure and network stability of the plankton community in multi-tributary river basins. *Journal of Hydrology*, 643, 131945. (IF=6.3, JCR Q1)
7. **Chang, C.**, Gao, L., Zamyadi, A., Wang, H., & Li, M.* (2025). Spatial dynamics of soil algae: Insights into abundance, community structure, and ecological roles in mixed biocrusts across China. *Applied Soil Ecology*, 208, 105974. (IF=5, JCR Q1)
8. **Chang, C.**, Gao, L., Zamyadi, A., Liu, S., Qu, N., & Li, M.* (2025). Temperature-driven deterministic assembly processes facilitate the coexistence of photoautotrophic communities in greenfield biocrusts across China. *Plant and Soil*, 1-20. (IF=4.1, JCR Q1)
9. **Chang, C.**, Hu, E., Shi, Y., Pan, B., & Li, M.* (2024). Linking microbial community coalescence to ecological diversity, community assembly and species coexistence in a typical subhumid river catchment in northern China. *Science of The Total Environment*, 938, 173367. (IF=8)
10. **Chang, C.**, Gao, L., Wei, J., Ma, N., He, Q., Pan, B., & Li, M. (2021). Spatial and environmental factors contributing to phytoplankton biogeography and biodiversity in mountain ponds across a large geographic area. *Aquatic Ecology*, 55(2), 721-735. (IF=1.8, JCR Q2)
11. Chen, G., **Chang, C.** (co-first author), He, X., Zhang, Q., Li, M., Wang, J., Geng, R., Qiao, J., Zhang, F., & Wang, H.* (2024). Variations in microbial characteristics of overland flow from steep slopes with biocrusts during rainfall in a semiarid region. *Journal of Hydrology*, 635, 131158. (IF=6.3, JCR Q1)
12. Chen, G., **Chang, C.** (co-first author), Yan, Z., Wang, H.*, Zhang, Q., Wang, J., & Zhu, B. (2026). Cross-domain microbial interactions mediate soil multifunctionality during crop growth stages. *Applied Soil Ecology*, 222, 106992. (IF=5, JCR Q1)

(b) Other author

13. Ye, S., Gao, L., **Chang, C.**, Zhang, X., Zhou, Y., Xiao, M., & Li, M.* (2026). Ranking distribution reveals opposite shifts in evenness and survival thresholds of phytoplankton under environmental stress. *The ISME Journal*, 20(1), wrag049.
14. Ye, S., **Chang, C.**, Xiao, M., Li, B., Hu, E., & Li, M.* (2025). Water travel time shapes the biogeography of riverine phytoplankton under the influence of dam and tributary. *Limnology and Oceanography*, 70(12), 3576-3591.

15. Ye, S., Wen, H., **Chang, C.**, Yang, F., Ma, Y., Tang, X., & Li, M.* (2025). River network size-dependent chemodiversity of dissolved organic matter and its driving factors in a multi-tributary mountain river. *Water Research*, 124580.
16. Yang, F., **Chang, C.**, Wen, L., Hao, Z., Pang, Y., Yi, Y., He, D., Li, M.*, 2025. Unravelling riverine dissolved organic matter sources using molecular fingerprints and FEAST model in a multi-tributary mountain river basin. *Water Research*, 123478.
17. Chen, G., Wang, H.*, Zhang, Q., **Chang, C.**, Wang, F., Zhang, Q., Wang, J., Pang, Y., Li, M., Zhu, B., He, D., 2025. Unraveling the links between molecular weight and multidimensional dynamics of dissolved organic matter in the runoff from gentle sloping land under rainfall. *Water Research*, 124061.
18. Lakshmikandan, M., Li, B., Tian, Y., Liu, Y., Yang, Y., Ye, S., Wen, H., **Chang, C.**, Li, M.*, 2025. Biodegradation of dimethyl phthalate by *Coelastrella terrestris* MLUN1: Metabolic adaptations and biodiesel potential. *Process Safety and Environmental Protection* 202, 107548.
19. Tang, X., Zhang, C., Xu, S., Hu, E., **Chang, C.**, Ma, Y., Xi, D., Li, M.*, 2025. Spatial and temporal dynamics of the complexity, stability and community assembly of phytoplankton community in a deep channel-type reservoir. *Journal of Environmental Sciences*.
20. Ye, S., Xu, S., Ren, M., **Chang, C.**, Hu, E., Li, M.*, 2025. Land use types, basin characteristics and water quality together shape riverine phytoplankton community composition and diversity. *Journal of Environmental Management* 376, 124496.
21. Tang, X., **Chang, C.**, Li, M.*, 2025. Pyrolysis gas chromatography mass spectrometry (py-GC/MS) revealed independent effects of precipitation and temperature on diversity and molecular composition of soil organic matter. *Environmental Earth Sciences* 84(6), 164.
22. Li, Y., Chen, Q., **Chang, C.**, Zhang, M., Li, M., Cui, L.*, 2025. Biogeographical patterns of antibiotic resistomes in soil biocrusts from urban green spaces at the continental scale. *Applied Soil Ecology* 212, 106150.
23. Qiao, R., Tian, Y., Wang, J., Lakshmikandan, M., Liu, Y., **Chang, C.**, Li, M.*, 2025. Nutrient removal from the effluent of wastewater treatment plants by eight newly isolated green microalgal species and the potential utilization of pyrolysis products from microalgal biomass. *Biomass and Bioenergy* 199, 107906.
24. Chen, G., Wang, H., **Chang, C.**, Zhang, Q., Wang, J., Zhang, F., Li, M.*, 2024. Migration pattern of protistan groups in the runoff from slopes covered with biocrusts under rainfall in a semiarid region. *Catena* 243, 108151.
25. Lakshmikandan, M., Yang, F., Ye, S., Liu, Y., **Chang, C.**, Yang, Y., Wen, H., Ameen, F., Li, M.*, 2024. Enhancing nutrient removal of agricultural and agro-industrial wastewater utilizing symbiotic microalgal co-cultivation systems to optimize sustainable resource recovery. *Journal of Water Process Engineering* 63, 105524.
26. Li, B., **Chang, C.**, Sun, C., Zhao, D., Hu, E., Li, M.*, 2024. Multi-habitat distribution and coalescence of resistomes at the watershed scale based on metagenomics. *Journal of Hazardous Materials* 477, 135349.
27. Liu, S., Fan, L., **Chang, C.**, Ge, Z., Ma, N., Chen, W., Yang, F., Pan, B., Li, M.*, Gao, L., 2022. Independent response of soil DOM to MAT and MAP: Evidence from a large-scale survey of moss crusts in mainland China. *Ecological Indicators* 142, 109240.
28. He, H.*, Wu, M., Su, R., Zhang, Z., **Chang, C.**, Peng, Q., Dong, Z., Pang, J., Lambers, H., 2021. Strong phosphorus (P)-zinc (Zn) interactions in a calcareous soil-alfalfa system suggest that rational P fertilization should be considered for Zn biofortification on Zn-deficient soils and phytoremediation of Zn-contaminated soils. *Plant and Soil* 461, 119-134.
29. He, H.*, Zhang, Z., Peng, Q., **Chang, C.**, Su, R., Cheng, X., Li, Y., Pang, J., Du, S., Lambers, H., 2021. Increasing nitrogen supply to phosphorus-deficient *Medicago sativa* decreases shoot growth and enhances root exudation of tartrate to discharge surplus carbon dependent on nitrogen form. *Plant and Soil* 469, 193-211.
30. Zhang, Z., Su, R., **Chang, C.**, Cheng, X., Peng, Q., Lambers, H., He, H.*, 2021. Effects of oxytetracycline on plant growth, phosphorus uptake, and carboxylates in the rhizosheath of alfalfa. *Plant and Soil* 461, 501-515.

Honors and Awards

Graduate first-class scholarship / second-class scholarship, Northwest A&F University, 2019-2024

Research Experience

- Youth project of the national natural science foundation of China: the impact of polysaccharide structure and composition on the formation of *Microcystis* populations and phenotypic transformation.
- Shaanxi provincial natural science foundation project: study on the impact of water level changes in the Hancheng-Weihe reservoir area on wetland ecosystems and countermeasures.
- Shaanxi provincial water conservancy technology project: study on the ecological evolution of water sources in Loess Plateau reservoirs and regulation strategies.
- Cyrus Tang Foundation Sponsored Project: research on the deep treatment of microalgae for the reuse of urban and

rural domestic wastewater in irrigation fields.

Sampling Experience

- Led a team from March to November 2023 to collect environmental samples, including river, soil, and other samples, from 145 sampling points across 24 main and tributary rivers in the upper Hanjiang River and Weihe River Basin in the Qinba Mountain Area. Responsible for planning the sampling point layout, coordinating sample collection, lab analysis, and subsequent data analysis.
- Led a team in June 2021 to collect inflow, outflow, and sludge samples from over 50 wastewater treatment plants in Shaanxi Province.
- Led a team in April and July-August 2021 to collect samples from 15 rivers in the Shaanxi Province.
- Additionally, participated in multiple field sampling activities from 2021 to 2024, gaining extensive experience in field sample collection.

Professional Skills

- Proficient in using analytical instruments such as TOC analyzer, 3D fluorescence spectrometer, ion chromatograph, Kjeldahl nitrogen analyzer, and UV-Vis spectrophotometer.
- Proficient in R programming and Linux operating system, capable of independently completing amplicon and metagenomic data analysis. I have processed over 5,000 amplicon samples and 500 metagenomic samples. Currently studying transcriptomics and metabolomics. Able to perform personalized bioinformatics analyses and possess foundational knowledge in microbial ecology and community ecology. Skilled in using Microsoft Office software, AI for drawing, Origin, and other related software.

Professional Qualifications

- Serve as an independent reviewer for international academic journals such as ***Environment International***, ***Journal of Environmental Management***, ***Applied Soil Ecology***, and ***Process Safety and Environmental Protection***.